

25 Research-Grade Engineering Capstone Projects

For a 4-student team · 7-month timeline · moderate budget (cloud GPUs allowed) · undergraduate CSE

Each idea below is scoped to require a genuine original contribution (new algorithm/protocol/architecture), not integration of off-the-shelf components. Ratings are on a 1-10 scale unless noted.

1. FloodMesh — Opportunistic LoRa-Ultrasonic Hybrid Mesh for Disaster Communication

Problem Statement: After earthquakes/floods, cellular and Wi-Fi infrastructure collapses, leaving first responders without reliable local communication for hours to days.

Why existing solutions fail: Pure LoRa mesh (e.g., Meshtastic-style) has poor throughput and breaks down in dense rubble/urban canyons due to RF shadowing; pure acoustic/ultrasonic systems have very short range and high power draw. No system adaptively switches modality based on real-time link quality.

Research Gap: No published routing protocol jointly optimizes over two heterogeneous physical layers (sub-GHz RF + ultrasonic acoustic) with dynamic modality switching based on instantaneous channel state.

Proposed Innovation: A **Dual-Modality Adaptive Routing Protocol (DMARP)** that treats each node's RF and acoustic links as separate weighted graph edges and reroutes packets hop-by-hop using a modality-cost function updated via lightweight channel probing.

Technical Architecture: ESP32 + LoRa SX1276 module + piezo ultrasonic transducer per node; local controller runs DMARP routing daemon; gateway node bridges to a disaster-ops dashboard over satellite uplink when available.

Algorithms: New DMARP: a modified Bellman-Ford using a composite cost metric (latency, RSSI/acoustic-SNR, battery) with hysteresis to avoid route flapping; store-and-forward buffering for disconnection tolerance (DTN-style).

Technologies: LoRa (SX1276), ultrasonic transducers (40 kHz), ESP-IDF, Zephyr RTOS optional, MQTT-SN for gateway bridging.

Dataset: Self-collected field-trial RSSI/acoustic-SNR traces in simulated rubble (indoor obstruction course); optionally augment with public LoRa urban-canyon RSSI datasets.

Hardware: 8-12 ESP32 dev boards, LoRa modules, ultrasonic transducers, batteries (~\$40/node); no GPUs required.

Expected measurable improvement: Target $\geq 30\%$ improvement in packet delivery ratio and $\geq 25\%$ reduction in end-to-end latency vs. single-modality LoRa mesh baseline (Meshtastic) under obstructed test course.

Evaluation Metrics: Packet delivery ratio, end-to-end latency, energy per delivered byte, route stability (flap rate).

Difficulty: 6/10 **Research Novelty:** 7/10 **Implementation Complexity:** Medium **IEEE Publication Potential:** High (IEEE Sensors, INFOCOM workshop, or ICC disaster-networking track) **Patent Potential:** Medium (protocol-level claim on modality-switching heuristic) **Industry Demand:** High (disaster response, defense, mining, off-grid IoT) **Estimated Timeline:** M1–2 hardware bring-up & single-modality baselines; M3–4 DMARP design + simulation; M5–6 field trials; M7 writeup. **Why judges will find it impressive:** Combines two physical layers no one has jointly optimized, has a physical rubble-course demo, and solves a life-safety problem with a genuinely new routing algorithm.

2. FlickerFree-LiFi — Adaptive Constellation VLC Protocol for Flicker-Free Gigabit LiFi

Problem Statement: Visible Light Communication (LiFi) must maintain constant perceived brightness (flicker-free, dimmable) while maximizing throughput, but current modulation schemes (OOK, standard OFDM-VLC) trade off badly between dimming range and data rate.

Why existing solutions fail: Variable Pulse Position Modulation and standard DC-biased optical OFDM either sacrifice throughput at low dimming levels or introduce visible flicker; IEEE 802.15.7 reference schemes are not adaptive to real-time channel/dimming conditions.

Research Gap: No adaptive modulation scheme jointly optimizes constellation size and DC-bias per OFDM subcarrier as a function of instantaneous dimming target and channel SNR, with formal flicker constraints.

Proposed Innovation: Adaptive-Bias Bit-and-Power-Loaded DCO-OFDM (AB-BPL) — a new bit/power loading algorithm that solves a constrained optimization (max throughput s.t. flicker index < threshold and dimming level = target) per-frame using a fast greedy water-filling variant.

Technical Architecture: LED transmitter driven by DAC + custom analog front-end; photodiode receiver with TIA and ADC; FPGA or Raspberry Pi + SDR-like baseband processing for real-time OFDM.

Algorithms: New constrained bit-loading algorithm (modified Chow/Campello loading with dimming-aware Lagrangian); real-time flicker-index estimator.

Technologies: MATLAB/Simulink for algorithm design, GNU Radio-style SDR pipeline on FPGA (e.g., Zynq) or Raspberry Pi + ADC/DAC HAT, C/C++ for embedded DSP.

Dataset: Self-collected: throughput vs. dimming-level vs. distance measurements from custom testbed (no public dataset exists for this specific setup).

Hardware: High-speed LED + driver circuit, PIN photodiode + transimpedance amp, FPGA dev board (~\$200), oscilloscope for validation.

Expected measurable improvement: $\geq 40\%$ higher throughput at 20–80% dimming levels vs. standard DCO-OFDM baseline, while keeping flicker index below IEEE 1789 recommended limits.

Evaluation Metrics: Throughput (Mbps) vs. dimming level curve, flicker index, BER, achievable dimming range.

Difficulty: 8/10 **Research Novelty:** 8/10 **Implementation Complexity:** High (analog + FPGA DSP) **IEEE Publication Potential:** High (IEEE Photonics Technology Letters, GLOBECOM, ICC) **Patent Potential:** High (novel modulation/loading algorithm) **Industry Demand:** Medium-High (smart lighting, indoor connectivity, aviation) **Estimated Timeline:** M1-2 analog front-end + baseline OOK/OFDM; M3-5 AB-BPL algorithm + FPGA implementation; M6 testbed characterization; M7 writeup. **Why judges will find it impressive:** Solves a real, quantifiable engineering tradeoff (light quality vs. bandwidth) with a novel constrained-optimization algorithm and a working optical link demo.

3. FedCluster-NC — Non-IID-Aware Federated Learning via Online Client Clustering with Gradient-Conflict Detection

Problem Statement: Federated learning degrades sharply when client data distributions are highly non-IID (e.g., hospitals with different patient demographics), causing slow/unstable convergence.

Why existing solutions fail: FedAvg assumes roughly homogeneous data; clustered FL methods (e.g., IFCA) use static or coarse clustering that doesn't adapt to evolving gradient conflict patterns across training rounds, and don't scale cluster count dynamically.

Research Gap: No method dynamically detects gradient-conflict subgroups online (without needing client identity/metadata) and reallocates clients to clusters using a principled, provably convergent update rule.

Proposed Innovation: Online Gradient-Conflict Clustering (OGCC) — a new server-side algorithm that computes pairwise gradient cosine-conflict scores from compressed gradient sketches (to preserve privacy/bandwidth), forms clusters via streaming spectral clustering, and aggregates within-cluster using a conflict-weighted FedAvg variant.

Technical Architecture: Central server orchestrates rounds; clients run local SGD; server maintains a sketch-based conflict graph updated incrementally; new aggregation module replaces standard FedAvg.

Algorithms: Streaming spectral clustering on compressed gradient sketches (count-sketch dimensionality reduction), conflict-weighted aggregation rule with convergence proof sketch, adaptive cluster-count selection via eigengap heuristic.

Technologies: PyTorch + Flower or custom federated simulator, NumPy/SciPy for spectral methods, optional differential privacy noise injection.

Dataset: FEMNIST, CIFAR-100 non-IID partitions (Dirichlet-based), and a synthetic multi-hospital medical imaging benchmark (e.g., partitioned ChestX-ray14) for a realistic non-IID case study.

Hardware: Cloud GPU (1× A100 or 4× T4 equivalent) for simulating up to 100 clients.

Expected measurable improvement: ≥ 15 –20% higher final accuracy and $\geq 2\times$ faster convergence (rounds to target accuracy) vs. FedAvg and IFCA baselines under high non-IID skew (Dirichlet $\alpha = 0.1$).

Evaluation Metrics: Test accuracy vs. rounds, communication cost, cluster purity vs. ground-truth domain labels, convergence rounds.

Difficulty: 7/10 **Research Novelty:** 8/10 **Implementation Complexity:** Medium-High **IEEE Publication Potential:** High (ICML/NeurIPS workshop, IEEE TPDS, INFOCOM) **Patent Potential:** Medium **Industry Demand:** High (healthcare AI, mobile keyboards, cross-silo FL) **Estimated Timeline:** M1-2 baseline FedAvg/IFCA reproduction; M3-4 OGCC design; M5-6 large-scale experiments; M7 writeup. **Why judges will find it impressive:** Tackles a well-known open FL problem with a principled online algorithm and rigorous ablations against strong recent baselines.

4. EvictAttn — Learned Importance-Aware KV-Cache Eviction for Memory-Efficient LLM Inference

Problem Statement: Serving long-context LLMs on memory-constrained (edge/single-GPU) hardware is bottlenecked by KV-cache size, which grows linearly with sequence length.

Why existing solutions fail: Heuristic eviction methods (H2O, StreamingLLM) use fixed attention-score heuristics that don't adapt per-task and can evict tokens needed for long-range reasoning, causing accuracy cliffs on retrieval-heavy tasks.

Research Gap: No lightweight, learned eviction policy that predicts *future* token importance (not just past attention mass) using a small auxiliary predictor trained jointly with minimal overhead.

Proposed Innovation: Predictive Importance Eviction (PIE) — a tiny auxiliary MLP head trained to predict each token's expected future attention contribution, used to score cache entries for eviction under a fixed memory budget, combined with a segment-recency prior for stability.

Technical Architecture: Wraps around any decoder-only transformer (e.g., LLaMA-family) inference loop; auxiliary predictor runs alongside attention computation with negligible added FLOPs; custom CUDA kernel for fused score-and-evict.

Algorithms: New eviction scoring function combining learned importance + recency decay; online budgeted top-k cache management (new data structure: a priority-heap-backed sliding cache).

Technologies: PyTorch, HuggingFace Transformers, custom CUDA/Triton kernels, vLLM or llama.cpp integration for realistic serving benchmarks.

Dataset: LongBench, RULER (long-context retrieval benchmark), PG19 for perplexity evaluation.

Hardware: 1-2 cloud GPUs (A100/L4) for training the auxiliary predictor and running inference benchmarks.

Expected measurable improvement: $\geq 40\%$ KV-cache memory reduction at $\leq 2\%$ accuracy drop on long-context QA/retrieval tasks vs. H2O/StreamingLLM at matched memory budgets.

Evaluation Metrics: Memory footprint, perplexity, retrieval accuracy on RULER, tokens/sec throughput.

Difficulty: 8/10 **Research Novelty:** 8/10 **Implementation Complexity:** High (CUDA kernel work) **IEEE Publication Potential:** High (MLSys, ACL, NeurIPS efficiency workshop) **Patent Potential:** Medium-High **Industry Demand:** Very High (LLM serving cost is a top industry pain point) **Estimated Timeline:** M1-2 baseline reproduction (H2O/StreamingLLM); M3-5 PIE design + kernel implementation; M6 benchmarking on LongBench/RULER; M7 writeup. **Why judges will find it impressive:** Directly attacks the #1 cost driver in LLM serving with a learned (not heuristic) method and real kernel-level engineering.

5. LEO-CC — Orbit-Aware Congestion Control for Low-Earth-Orbit Satellite Internet

Problem Statement: Standard TCP congestion control (Cubic, BBR) misinterprets the periodic latency/bandwidth variations caused by LEO satellite handovers as congestion, causing unnecessary throughput collapse every few minutes.

Why existing solutions fail: BBR and Cubic are reactive and handover-agnostic; they treat orbit-induced RTT spikes identically to terrestrial congestion, and don't use publicly available ephemeris data to anticipate handovers.

Research Gap: No congestion control algorithm incorporates satellite ephemeris/orbital prediction as a first-class signal to pre-emptively adjust the congestion window before a predicted handover.

Proposed Innovation: Ephemeris-Predictive Congestion Control (EPCC) — augments a BBR-style bandwidth/RTT model with a handover predictor (using TLE orbital data) that proactively widens the pacing gain window just before predicted handovers and tightens it during stable windows to avoid false congestion signals.

Technical Architecture: Linux kernel congestion-control module (pluggable CCA) + userspace ephemeris predictor feeding hints via a lightweight sysfs/netlink interface.

Algorithms: New hybrid CCA combining BBRv2-style bandwidth estimation with a handover-aware state machine; SGP4-based orbital propagation for handover time prediction.

Technologies: Linux TCP CCA kernel module (C), Mininet/ns-3 for simulation, real Starlink RTT/throughput traces for validation, Python (skyfield/sgp4) for ephemeris prediction.

Dataset: Public Starlink measurement datasets (e.g., academic Starlink RTT/throughput traces), public TLE data from CelesTrak.

Hardware: No special hardware; optional access to a Starlink terminal for live validation (nice-to-have, not required).

Expected measurable improvement: $\geq 25\%$ higher sustained throughput and $\geq 50\%$ reduction in throughput collapse events during handovers vs. Cubic/BBR baselines in ns-3 LEO simulation.

Evaluation Metrics: Throughput, RTT variance, goodput during handover windows, fairness index (Jain's).

Difficulty: 8/10 **Research Novelty:** 7/10 **Implementation Complexity:** High (kernel-level networking) **IEEE Publication Potential:** High (SIGCOMM workshop, INFOCOM, IEEE/ACM ToN) **Patent Potential:** Medium-High **Industry Demand:** Very High (Starlink, Kuiper, OneWeb ecosystem) **Estimated Timeline:** M1-2 ns-3 LEO simulation setup + baseline CCAs; M3-5 EPCC design and kernel/simulation implementation; M6 trace-driven validation; M7 writeup. **Why judges will find it impressive:** Addresses a real, current problem in the fast-growing LEO internet industry with a concrete, testable kernel-level contribution.

6. LightPQ — Lattice-Based Lightweight Key Exchange for Resource-Constrained IoT

Problem Statement: IoT devices need post-quantum-secure key exchange, but standard PQC schemes (Kyber, NTRU) are too compute/memory-heavy for microcontrollers with <64KB RAM.

Why existing solutions fail: NIST PQC finalists are designed for general-purpose CPUs; existing "lightweight" adaptations either cut security margins unsafely or don't meaningfully reduce RAM/energy footprint for 8/32-bit MCUs.

Research Gap: No key-exchange protocol co-designs a reduced-parameter lattice scheme with a hardware-aware polynomial multiplication algorithm optimized for MCU-class SIMD-less architectures, with a formal security-vs-footprint tradeoff analysis.

Proposed Innovation: MicroKEX — a parameterized lattice-based KEX using a memory-streaming Number-Theoretic Transform (NTT) variant that processes polynomial coefficients in fixed-size chunks (never materializing the full polynomial in RAM), plus a new parameter-selection algorithm that certifies minimum security level for a given RAM budget.

Technical Architecture: Runs on ARM Cortex-M0/M4 class MCUs; streaming-NTT library in C; integrated into a DTLS-like handshake for constrained networks (6LoWPAN/CoAP).

Algorithms: New chunked/streaming NTT algorithm with bounded scratch memory; new security-parameter search algorithm balancing lattice dimension vs. RAM ceiling.

Technologies: C (bare-metal), ARM Cortex-M dev boards, CoAP/6LoWPAN stack, formal analysis via the Lattice Estimator tool for security verification.

Dataset: Not applicable (protocol/systems work); benchmark against NIST PQC reference implementations.

Hardware: STM32 or nRF52 Cortex-M dev boards (~\$15-30 each), power measurement rig (e.g., Joulescope or INA219-based).

Expected measurable improvement: $\geq 50\%$ RAM reduction and $\geq 30\%$ energy-per-handshake reduction vs. reference Kyber-512 implementation on Cortex-M0, at equivalent NIST security level 1.

Evaluation Metrics: Peak RAM usage, handshake latency, energy per handshake, formal security level (bits).

Difficulty: 9/10 **Research Novelty:** 8/10 **Implementation Complexity:** High (crypto engineering + formal security analysis) **IEEE Publication Potential:** High (IEEE IoT Journal, CHES workshop, ACM CCS poster) **Patent Potential:** High **Industry Demand:** High (IoT security is a mandated, growing requirement) **Estimated Timeline:** M1-2 study PQC + reference implementations; M3-5 streaming-NTT + protocol design; M6 MCU benchmarking + security validation; M7 writeup. **Why judges will find it impressive:** Combines applied cryptography with embedded systems engineering and produces a certifiably secure, measurably lighter protocol — rare undergraduate-level rigor.

7. SwarmLoc — GPS-Denied Cooperative Localization for Drone Swarms via Ultra-Wideband Ranging and Distributed Consensus

Problem Statement: Drone swarms operating indoors, underground, or in GPS-jammed environments (disaster search, mining) cannot localize themselves accurately using GPS, and existing SLAM approaches don't scale well to many cooperating agents in real time.

Why existing solutions fail: Single-agent visual/LiDAR SLAM is compute-heavy for small drones and drifts over time; existing cooperative localization methods assume reliable all-to-all communication, which fails in cluttered environments with intermittent UWB/radio links.

Research Gap: No distributed localization algorithm combines UWB ranging with a robust consensus protocol that tolerates partial/asymmetric connectivity and byzantine/faulty range measurements without a central coordinator.

Proposed Innovation: Robust Distributed Consensus Localization (RDCL) — a new gossip-based consensus algorithm where each drone maintains a local position belief updated via a robust (Huber-loss) weighted least-squares fusion of UWB ranges and IMU dead-reckoning, with outlier range rejection via a distributed RANSAC variant.

Technical Architecture: Each drone has UWB module (DW1000) + IMU; onboard companion computer (e.g., Raspberry Pi/Jetson Nano) runs RDCL; drones exchange belief states opportunistically over mesh radio.

Algorithms: New robust gossip-consensus localization algorithm with Huber-loss fusion; distributed outlier-rejection (RANSAC-over-gossip) for faulty range measurements; drift-correction via periodic anchor re-sync.

Technologies: ROS2, DW1000 UWB modules, PX4/ArduPilot for flight control, Python/C++ for the localization stack, Gazebo for simulation before real flights.

Dataset: Self-collected UWB ranging + IMU logs from testbed flights; UTIAS or similar public UWB-SLAM datasets for baseline comparison.

Hardware: 4-6 small quadrotors, DW1000 UWB modules (~\$25 each), Raspberry Pi/Jetson Nano companion computers, motion-capture system (if available) for ground truth, or fallback to fixed UWB anchors for ground truth.

Expected measurable improvement: $\geq 40\%$ reduction in localization RMSE vs. IMU-only dead reckoning, and graceful degradation ($\leq 15\%$ RMSE increase) under 30% simulated link-failure rate, vs. centralized baseline which fails outright.

Evaluation Metrics: Localization RMSE, convergence time after link disruption, robustness under simulated byzantine ranges, swarm scalability (up to N drones in simulation).

Difficulty: 9/10 **Research Novelty:** 8/10 **Implementation Complexity:** High (real hardware + distributed algorithms) **IEEE Publication Potential:** High (ICRA, IROS, IEEE RA-L) **Patent Potential:** Medium-High **Industry Demand:** High (search-and-rescue, warehouse robotics, mining, defense) **Estimated Timeline:** M1–2 simulation in Gazebo + algorithm design; M3–4 real hardware bring-up; M5–6 field trials with increasing swarm size; M7 writeup. **Why judges will find it impressive:** A live multi-drone demo with a genuinely novel distributed, fault-tolerant algorithm is a strong, visually compelling capstone centerpiece.

8. EdgeSched — Deadline-Aware Dynamic Offloading Scheduler for Real-Time Multi-Stream Computer Vision at the Edge

Problem Statement: Edge devices (e.g., smart cameras) running multiple concurrent CV pipelines (detection, tracking, segmentation) must decide, per frame and per stream, whether to run locally or offload to a nearby edge server — under tight latency deadlines and fluctuating network conditions.

Why existing solutions fail: Existing offloading frameworks (e.g., static partitioning, simple threshold-based offload) don't jointly optimize across multiple concurrent streams with heterogeneous deadlines, and don't adapt online to network jitter.

Research Gap: No scheduler formulates multi-stream edge/cloud CV offloading as an online constrained optimization solved with a lightweight learned policy that generalizes across unseen network conditions.

Proposed Innovation: Deadline-Constrained Offload Scheduler (DCOS) — a new scheduling algorithm combining a fast heuristic (Earliest-Deadline-First-with-Offload-cost) with a lightweight contextual-bandit policy that learns, per stream, the offload-vs-local decision boundary based on real-time network RTT/bandwidth estimates and model-accuracy/latency tradeoffs (via early-exit CV models).

Technical Architecture: Edge device (Jetson) hosts early-exit CV models locally; nearby edge server hosts full-accuracy models; scheduler sits between camera streams and inference, making per-frame local/offload decisions.

Algorithms: New deadline-aware scheduling heuristic + contextual bandit (LinUCB variant) for adaptive offload decisions; early-exit neural network design for tunable local latency/accuracy tradeoff.

Technologies: NVIDIA Jetson Orin/Nano, PyTorch with early-exit branches (e.g., modified YOLO/ResNet), gRPC for edge-server offload, network emulation via $\overline{\text{tc}}$ /Mininet.

Dataset: COCO/VisDrone for detection accuracy; self-collected multi-stream latency/network traces from testbed.

Hardware: 1-2 Jetson Orin Nano boards, a local "edge server" (workstation with GPU), Wi-Fi router with controllable bandwidth shaping.

Expected measurable improvement: $\geq 20\%$ reduction in deadline-miss rate and $\geq 10\%$ higher average detection accuracy (via smarter offload use) vs. static-threshold and greedy-always-offload baselines, under realistic jittery network traces.

Evaluation Metrics: Deadline-miss rate, end-to-end latency distribution, mAP (detection accuracy), offload decision regret vs. oracle.

Difficulty: 7/10 **Research Novelty:** 7/10 **Implementation Complexity:** Medium-High **IEEE Publication Potential:** High (IEEE Edge/Fog computing venues, MobiSys workshop, IoTJ) **Patent Potential:** Medium **Industry Demand:** High (smart city cameras, AR/VR, autonomous systems) **Estimated Timeline:** M1-2 early-exit model training + baseline schedulers; M3-5 DCOS bandit algorithm + integration; M6 multi-stream network-emulated evaluation; M7 writeup. **Why judges will find it impressive:** Combines systems scheduling theory, online learning, and efficient CV model design into one coherent, demoable pipeline.

9. SparseRV — A Compiler Pass for Automatic Sparse Tensor Kernel Generation on RISC-V Vector Extensions

Problem Statement: Sparse neural network inference (pruned models) on emerging RISC-V vector (RVV) hardware currently requires hand-written kernels; no compiler automatically generates efficient sparse-aware vectorized code for RVV targets.

Why existing solutions fail: Existing sparse compilers (e.g., TACO, SparseTIR) target x86/CUDA and don't account for RVV's variable vector-length (VLEN-agnostic) model or its specific gather/scatter instruction costs, leading to poor code when naively retargeted.

Research Gap: No compiler intermediate representation + lowering pass specifically models RVV's vector-length-agnostic execution for sparse formats (CSR/CSC/block-sparse), leaving significant performance on the table.

Proposed Innovation: RVV-Sparse Lowering Pass (RVSLP) — a new MLIR dialect and lowering pass that represents sparse tensor operations abstractly and generates VLEN-agnostic RVV assembly using a new cost model that predicts optimal blocking/tiling for a given sparsity pattern.

Technical Architecture: Built as an MLIR/LLVM compiler pass; sparse tensors described via a new IR dialect; lowering targets RVV intrinsics; tested on RVV-capable hardware or QEMU/gem5 RVV simulation.

Algorithms: New cost-model-driven tiling/blocking selection algorithm for sparse formats on VLEN-agnostic hardware; new sparse-to-dense-blocking heuristic tuned via small autotuning search.

Technologies: MLIR/LLVM, RISC-V GNU toolchain, QEMU with RVV support or SiFive/gem5 simulators, Python for autotuning harness.

Dataset: Standard sparse NN benchmark suite (pruned ResNet/BERT weight matrices at varying sparsity levels), SuiteSparse matrix collection for general sparse-kernel

benchmarking.

Hardware: RVV-capable dev board if available (e.g., SiFive boards) or QEMU/gem5 simulation (sufficient given budget constraints); cloud CPU time for compilation experiments.

Expected measurable improvement: $\geq 1.5\text{--}2\times$ speedup on sparse matrix-vector/matrix-matrix kernels vs. naive RVV-retargeted TACO-generated code, across 50–90% sparsity range.

Evaluation Metrics: Kernel runtime (cycles), compile time, code size, speedup vs. dense baseline and vs. hand-tuned reference kernels where available.

Difficulty: 9/10 **Research Novelty:** 8/10 **Implementation Complexity:** Very High (compiler internals) **IEEE Publication Potential:** High (CGO, PACT, MLSys) **Patent Potential:**

Medium **Industry Demand:** High (RISC-V is a major growth area in AI hardware)

Estimated Timeline: M1–3 MLIR/RVV toolchain ramp-up + IR design; M4–6 lowering pass + cost model + autotuning; M7 evaluation + writeup. **Why judges will find it impressive:** Compiler research is rare at the undergraduate capstone level; a working, measurable speedup on real emerging hardware is a strong differentiator.

10. HAOS-Sched — A Heterogeneity-Aware OS Kernel Scheduler for Mixed CPU/NPU/GPU Edge AI Workloads

Problem Statement: Modern edge SoCs (Jetson, Snapdragon, Apple-silicon-class) have CPU, GPU, and NPU accelerators, but the Linux CFS scheduler has no notion of accelerator contention, causing poor latency/throughput when multiple AI tasks compete for the same NPU/GPU.

Why existing solutions fail: CFS schedules CPU time only; accelerator access is typically FIFO or driver-default, with no fairness or priority awareness across processes, causing head-of-line blocking for latency-critical tasks (e.g., safety-critical perception) behind throughput-oriented batch jobs.

Research Gap: No published Linux scheduler extension provides a unified, priority- and deadline-aware scheduling policy that spans CPU cores *and* accelerator queues jointly, with formal starvation-freedom guarantees.

Proposed Innovation: Cross-Domain Deadline Scheduler (XDDS) — a new Linux kernel scheduling class that extends CFS/EDF concepts to accelerator command queues via a kernel-level accelerator-queue shim, using a new co-scheduling algorithm that jointly assigns CPU time slices and accelerator queue priority to respect end-to-end task deadlines.

Technical Architecture: Linux kernel module implementing a new `sched_class`; accelerator queue interception via kernel driver shims (for NPU/GPU submission queues); userspace task-deadline annotation API.

Algorithms: New joint CPU+accelerator co-scheduling algorithm (extended EDF with resource-reservation across two resource classes); starvation-freedom proof via classical scheduling theory extension.

Technologies: Linux kernel (C), NVIDIA Jetson Orin (GPU/DLA queues) as target platform, `(perf)`/`(ftrace)` for measurement, custom benchmark task suite (mixed CV inference workloads).

Dataset: Not applicable (systems work); benchmark workloads: YOLO inference, image classification, and background batch training tasks.

Hardware: 1–2 Jetson Orin dev kits.

Expected measurable improvement: $\geq 30\%$ reduction in 99th-percentile latency for high-priority tasks under contention, with $\leq 5\%$ total system throughput loss, vs. stock Linux scheduler + default driver queuing.

Evaluation Metrics: Tail latency (p50/p95/p99), throughput, fairness (Jain's index), deadline-miss rate.

Difficulty: 9/10 **Research Novelty:** 8/10 **Implementation Complexity:** Very High (kernel development) **IEEE Publication Potential:** High (RTSS, EuroSys workshop, RTAS) **Patent Potential:** Medium-High **Industry Demand:** High (every edge-AI vendor has this exact pain point) **Estimated Timeline:** M1–3 kernel scheduler internals study + shim prototyping; M4–6 XDDS implementation + tuning; M7 benchmarking + writeup. **Why judges will find it impressive:** Kernel-level systems work is technically elite for undergrads and directly targets a documented industry pain point with measurable latency wins.

11. PerceptABR — Perceptual-Quality-and-Network-Prediction-Driven Adaptive Bitrate Streaming

Problem Statement: Adaptive bitrate (ABR) video streaming algorithms optimize for bitrate/rebuffering but ignore *perceptual* quality (e.g., content-dependent compressibility), leading to suboptimal QoE at a given bandwidth budget.

Why existing solutions fail: Buffer-based (BOLA) and throughput-based (RobustMPC) ABR algorithms treat all segments as equally "valuable" per bitrate level; learning-based ABR (Pensieve) optimizes a generic QoE reward but doesn't incorporate per-segment perceptual quality metrics (VMAF) predicted ahead of time without full encoding.

Research Gap: No ABR algorithm combines a lightweight per-segment VMAF *predictor* (avoiding costly multi-bitrate pre-encoding) with a network-throughput predictor in a joint decision policy, evaluated for real deployability (low compute overhead on client).

Proposed Innovation: Perceptual-Aware Predictive ABR (PAP-ABR) — a new reinforcement-learning-based ABR policy whose state includes a lightweight learned VMAF predictor (from encoded-frame complexity features, not full re-encoding) and an LSTM-based short-horizon throughput predictor, trained via a custom reward balancing perceptual quality, rebuffering, and switching smoothness.

Technical Architecture: Client-side ABR module (JS/Python) integrated with a streaming testbed (e.g., using DASH.js); server pre-computes lightweight complexity features per segment; RL policy runs on-device.

Algorithms: New lightweight VMAF-prediction regression model (trained offline, run online); RL policy (PPO) with novel reward function integrating predicted-VMAF;

throughput prediction via LSTM.

Technologies: Python, PyTorch (RL + VMAF predictor), DASH.js or custom streaming client, Mahimahi/ns-3 for realistic network trace replay.

Dataset: Public network throughput trace datasets (e.g., FCC broadband, HSDPA traces), public video test sets (e.g., UVG, Xiph) for VMAF ground truth.

Hardware: No special hardware; cloud GPU for RL training.

Expected measurable improvement: $\geq 15\%$ improvement in perceptual QoE (VMAF-weighted) at equal rebuffering rate vs. Pensieve and RobustMPC baselines across diverse network traces.

Evaluation Metrics: VMAF-weighted QoE score, rebuffering ratio, bitrate switch frequency, average bitrate.

Difficulty: 7/10 **Research Novelty:** 7/10 **Implementation Complexity:** Medium-High **IEEE Publication Potential:** High (ACM MMSys, NOSSDAV, INFOCOM) **Patent Potential:** Medium **Industry Demand:** High (every video streaming company cares about ABR QoE) **Estimated Timeline:** M1-2 baseline ABR reproduction (BOLA/Pensieve); M3-5 VMAF predictor + PAP-ABR RL policy; M6 trace-driven evaluation; M7 writeup. **Why judges will find it impressive:** Directly improves a metric (perceptual quality) that industry cares about more than raw bitrate, with a deployable, low-overhead design.

12. IntermitConsensus — A Consensus Protocol for Distributed Databases Under Intermittent Connectivity (Edge/Rural Deployments)

Problem Statement: Distributed databases deployed across edge sites with intermittent, high-latency links (rural clinics, ships, remote sensors) suffer from unavailability or inconsistency because classic consensus protocols (Raft/Paxos) require a stable quorum majority.

Why existing solutions fail: Raft/Paxos block writes when quorum is unreachable; CRDT-based eventual-consistency systems avoid blocking but give weak consistency guarantees unsuitable for applications needing bounded staleness with correctness guarantees (e.g., inventory, medical records).

Research Gap: No consensus protocol provides tunable, formally bounded staleness with adaptive quorum sizing based on predicted link-availability windows, rather than a fixed static majority requirement.

Proposed Innovation: Predictive Adaptive Quorum Consensus (PAQC) — a new consensus algorithm that adjusts effective quorum size dynamically using a link-availability predictor (based on historical connectivity patterns) and provides a formally bounded staleness guarantee via a hybrid logical-clock reconciliation mechanism when partitions heal.

Technical Architecture: Modified Raft implementation with a pluggable "quorum policy" module; link-availability predictor as a separate service; reconciliation module for post-partition merge.

Algorithms: New adaptive-quorum consensus algorithm (extension of Raft with dynamic quorum + staleness bound proof); Markov-chain-based link-availability predictor; hybrid logical clock (HLC) based conflict reconciliation.

Technologies: Go or Rust (for Raft implementation, e.g., based on `hashicorp/raft` or `etcd/raft` as a starting reference to extend, not copy application logic), ns-3/Mininet for network partition simulation.

Dataset: Simulated intermittent connectivity traces modeled from real rural/maritime satellite connectivity studies; synthetic workload generators (YCSB-style).

Hardware: Cloud VM cluster (5-7 nodes) for distributed testing; no GPU needed.

Expected measurable improvement: $\geq 95\%$ write availability under intermittent partition schedules (vs. $< 60\%$ for stock Raft) while maintaining formally bounded staleness ($\leq$ configurable threshold), validated via YCSB-style benchmarking.

Evaluation Metrics: Write availability, staleness bound violations, throughput, reconciliation conflict rate.

Difficulty: 8/10 **Research Novelty:** 7/10 **Implementation Complexity:** High (distributed systems correctness) **IEEE Publication Potential:** High (OSDI/EuroSys workshop, IEEE TPDS, ICDCS) **Patent Potential:** Medium **Industry Demand:** Medium-High (rural telemedicine, maritime, disaster-zone IT) **Estimated Timeline:** M1-2 Raft baseline + partition simulator; M3-5 PAQC algorithm + reconciliation design; M6 YCSB benchmarking; M7 writeup + correctness proof polish. **Why judges will find it impressive:** Rigorous distributed-systems theory (with a staleness-bound proof) applied to an underserved, socially valuable deployment scenario.

13. NeuroDenoise — A Novel Attention-Based Architecture for Real-Time EEG Denoising in Motor-Imagery BCIs

Problem Statement: Motor-imagery brain-computer interfaces suffer poor real-world accuracy because EEG signals are heavily contaminated by muscle/eye-movement artifacts, and current denoising methods either remove too much signal or run too slowly for real-time BCI control.

Why existing solutions fail: ICA-based artifact removal is not real-time-capable and requires manual component selection; existing deep denoising autoencoders are trained with generic reconstruction loss (MSE) that doesn't preserve the specific spectral bands (μ /beta) needed for motor-imagery classification.

Research Gap: No denoising architecture uses a classification-aware loss function that explicitly preserves task-relevant frequency bands while removing artifacts, in a causal (real-time-compatible) architecture.

Proposed Innovation: Band-Preserving Causal Denoising Network (BPC-DNet) — a new causal temporal-convolutional architecture with a custom **Spectral-Preservation Loss** (a new loss function combining time-domain reconstruction with a differentiable STFT-based penalty that specifically protects μ /beta band power) trained jointly with a downstream motor-imagery classifier in a multi-task setup.

Technical Architecture: Causal TCN denoiser → lightweight EEGNet-style classifier, trained end-to-end; real-time inference pipeline on a laptop/edge device processing streaming EEG.

Algorithms: New Spectral-Preservation Loss (differentiable STFT-band-power term); causal dilated-TCN denoising architecture; joint multi-task training schedule (denoising + classification).

Technologies: PyTorch, MNE-Python for EEG preprocessing, OpenBCI or similar consumer EEG headset for live data collection, LSL (Lab Streaming Layer) for real-time streaming.

Dataset: BCI Competition IV dataset 2a/2b (motor imagery) for offline training; self-collected pilot data with OpenBCI for real-time validation.

Hardware: Consumer EEG headset (OpenBCI Cyton, ~\$500), laptop for real-time inference (no GPU strictly required, but useful for training).

Expected measurable improvement: ≥ 10 –15% improvement in motor-imagery classification accuracy vs. raw-signal and standard-autoencoder-denoised baselines, with real-time latency under 50ms per window.

Evaluation Metrics: Classification accuracy (kappa score), signal-to-artifact ratio, mu/beta band power preservation, end-to-end latency.

Difficulty: 8/10 **Research Novelty:** 8/10 **Implementation Complexity:** High (signal processing + real-time systems) **IEEE Publication Potential:** High (IEEE TNSRE, EMBC, NER) **Patent Potential:** Medium **Industry Demand:** Medium-High (assistive tech, neurotech startups) **Estimated Timeline:** M1–2 dataset prep + baseline denoising/classification; M3–5 BPC-DNet + loss design; M6 real-time pilot data collection + validation; M7 writeup. **Why judges will find it impressive:** A live BCI demo (moving a cursor/robot via motor imagery) combined with a genuinely new, theoretically motivated loss function is highly compelling.

14. DP-FedMed — Adaptive Differential Privacy Budget Scheduling for Federated Medical Imaging

Problem Statement: Federated learning across hospitals for medical imaging must guarantee differential privacy (DP), but fixed per-round DP noise budgets either destroy model utility (too much noise early) or exhaust the privacy budget before convergence (too little noise control).

Why existing solutions fail: Standard DP-SGD/DP-FedAvg use a fixed noise multiplier across all rounds, ignoring the fact that gradient sensitivity and the marginal utility of privacy budget change over training — early rounds tolerate more noise, later fine-tuning rounds need less.

Research Gap: No published method adaptively schedules the per-round privacy budget allocation using an online estimate of the utility/privacy tradeoff curve, provably respecting a global (ϵ, δ) budget via privacy accountant composition.

Proposed Innovation: Utility-Adaptive Privacy Budget Scheduler (UAPBS) — a new algorithm that allocates the global DP budget non-uniformly across training rounds based on an online-estimated validation-utility-gain-per-epsilon curve, using a moment-accountant-compatible composition scheme to guarantee the total budget is respected.

Technical Architecture: Federated learning simulator (server + simulated hospital clients) with DP-SGD local training; central budget-scheduler module tracks cumulative privacy spend via a Rényi-DP accountant.

Algorithms: New adaptive budget-allocation algorithm (online convex optimization formulation); integration with Rényi differential privacy composition for formal guarantees.

Technologies: Opacus (PyTorch DP library) as the DP-SGD engine (extended, not just used off-the-shelf, since the scheduler itself is new), Flower or custom FL simulator.

Dataset: CheXpert or NIH ChestX-ray14, partitioned to simulate multiple hospital sites with realistic demographic skew.

Hardware: Cloud GPU (1× A100 or multiple T4s) for simulating 5–10 hospital clients.

Expected measurable improvement: ≥ 8 –12% higher final AUROC at the same total (ϵ, δ) privacy budget vs. fixed-noise DP-FedAvg baseline.

Evaluation Metrics: AUROC/accuracy at fixed privacy budget, privacy-utility tradeoff curve, formal (ϵ, δ) accounting verification.

Difficulty: 8/10 **Research Novelty:** 8/10 **Implementation Complexity:** High (requires solid grasp of DP theory) **IEEE Publication Potential:** High (IEEE JBHI, MICCAI workshop, PETS) **Patent Potential:** Medium **Industry Demand:** High (healthcare AI regulatory compliance is a major driver) **Estimated Timeline:** M1–2 DP-SGD/FedAvg baseline + privacy accountant setup; M3–5 UAPBS design + proof of budget correctness; M6 medical imaging experiments; M7 writeup. **Why judges will find it impressive:** Formal privacy guarantees plus measurable utility gains address a real regulatory/technical tension in healthcare AI that industry actively struggles with.

15. TinyStruct — Ultra-Low-Power Structural Health Monitoring via Learned Event-Triggered Compressive Sensing

Problem Statement: Continuous structural health monitoring (bridges, buildings) with battery/energy-harvested sensors is constrained by the energy cost of continuous vibration data sampling and transmission; existing systems either sample too sparsely (missing damage events) or drain batteries quickly.

Why existing solutions fail: Fixed-rate compressive sensing reduces data volume but still samples continuously regardless of structural activity; simple threshold-triggering misses gradual/subtle damage precursors that don't cross amplitude thresholds.

Research Gap: No system combines a learned, anomaly-aware trigger (detecting *subtle* precursor patterns, not just amplitude spikes) with an adaptive compressive-sensing rate that only increases fidelity when the trigger fires, under a strict energy budget.

Proposed Innovation: Learned Event-Triggered Adaptive Compressive Sensing (LETACS) — a new tiny (sub-10KB) on-device anomaly-scoring model that gates a

compressive-sensing sampling-rate controller, plus a new structured-random compressive sensing matrix design optimized jointly with the anomaly detector for reconstruction quality *only* on anomalous segments (compression algorithm contribution).

Technical Architecture: MEMS accelerometer + MCU (Cortex-M) node running LETACS; reconstructs signal server-side; energy-harvesting or coin-cell powered.

Algorithms: New tiny anomaly-detection model (quantized 1D-CNN or lightweight autoencoder, <10KB); new task-aware compressive sensing matrix design algorithm (optimized via a learned dictionary jointly with the anomaly detector, not generic random Gaussian matrices); adaptive sampling-rate controller.

Technologies: TensorFlow Lite Micro / CMSIS-NN, MEMS accelerometer (e.g., ADXL355), STM32 MCU, compressive sensing reconstruction (OMP/CoSaMP) server-side in Python.

Dataset: Public structural vibration datasets (e.g., Z24 Bridge benchmark, IASC-ASCE SHM benchmark) for offline training/validation; self-collected lab-scale vibration data (e.g., cantilever beam testbed) for real deployment validation.

Hardware: STM32 dev board, MEMS accelerometer, small shaker table/cantilever beam rig for controlled lab testing, power measurement tools.

Expected measurable improvement: $\geq 60\%$ energy reduction vs. fixed-rate continuous sampling baseline while maintaining $\geq 90\%$ damage-event detection recall, and $\geq 20\%$ better reconstruction SNR on anomalous segments vs. generic random compressive sensing matrices.

Evaluation Metrics: Energy consumption (mJ/day), detection recall/precision on labeled damage events, reconstruction SNR, device lifetime estimate.

Difficulty: 7/10 **Research Novelty:** 7/10 **Implementation Complexity:** Medium-High **IEEE**

Publication Potential: High (IEEE Sensors Journal, SHM conference, IoTJ) **Patent**

Potential: Medium **Industry Demand:** Medium-High (civil infrastructure monitoring is a growing IoT vertical) **Estimated Timeline:** M1-2 dataset study + baseline fixed-rate CS; M3-5 LETACS model + CS matrix co-design; M6 lab testbed validation; M7 writeup. **Why judges will find it impressive:** A physical shaker-table demo showing energy savings *and* better anomaly detection simultaneously is a compelling, tangible result.

16. EvacSwarm — Multi-Agent Reinforcement Learning for Adaptive Traffic Signal Control During Disaster Evacuation

Problem Statement: During mass evacuations (wildfires, floods), fixed-time or simple actuated traffic signals cause severe gridlock on evacuation routes, costing lives; existing adaptive traffic control isn't designed for the extreme, atypical demand surges of evacuation scenarios.

Why existing solutions fail: Standard adaptive signal control (SCOOT/SCATS, and most RL traffic-signal papers) is trained/tuned for typical daily traffic patterns and doesn't generalize to the extreme, spatially correlated demand surges of evacuations, nor does it coordinate explicitly toward evacuation-exit throughput as the objective.

Research Gap: No multi-agent RL traffic control method uses an evacuation-specific reward (network-wide time-to-clear) combined with a communication-efficient coordination mechanism that scales to city-scale intersection counts under emergency time constraints (training must generalize from limited disaster data).

Proposed Innovation: Evacuation-Aware Graph Multi-Agent RL (EA-GMARL) — a new multi-agent RL architecture where each intersection agent uses a graph-attention mechanism (new attention variant weighted by real-time downstream congestion propagation risk) to coordinate with neighbors, trained with a novel reward shaping term directly targeting network-wide evacuation clearance time rather than per-intersection wait time.

Technical Architecture: Trained in SUMO (traffic microsimulation) on real road network topology; graph-attention MARL policy (one shared policy, parameter-shared across intersections) deployed as a decentralized controller.

Algorithms: New "congestion-propagation-weighted" graph attention mechanism for inter-agent coordination; new evacuation-clearance-time reward shaping; MARL training via QMIX or MAPPO backbone extended with the new attention module.

Technologies: SUMO traffic simulator, PyTorch, PettingZoo/RLlib for MARL training, OpenStreetMap road network data for a real city's evacuation-relevant subnetwork.

Dataset: OpenStreetMap network topology for a real wildfire-prone region (e.g., a documented CA evacuation corridor), historical evacuation-order traffic studies for demand modeling, SUMO's built-in demand generation calibrated to those studies.

Hardware: Cloud GPU for MARL training (moderate; MARL training is CPU/simulation-heavy more than GPU-heavy, but useful for policy network training).

Expected measurable improvement: $\geq 25\%$ reduction in network-wide clearance time vs. fixed-time and standard actuated-signal baselines, and $\geq 15\%$ improvement vs. non-coordinated (independent) RL agents, in SUMO simulation of surge evacuation demand.

Evaluation Metrics: Network clearance time, average/max vehicle wait time, throughput at evacuation exits, scalability (number of intersections handled).

Difficulty: 8/10 **Research Novelty:** 7/10 **Implementation Complexity:** High (MARL + traffic simulation) **IEEE Publication Potential:** High (IEEE ITSC, T-ITS, AAMAS) **Patent Potential:** Low-Medium **Industry Demand:** Medium (municipal traffic management, disaster preparedness agencies) **Estimated Timeline:** M1-2 SUMO network setup + baseline signal controllers; M3-5 EA-GMARL design + training; M6 evacuation-scenario evaluation; M7 writeup. **Why judges will find it impressive:** A socially critical, life-safety application with a genuinely novel coordination mechanism and clear quantitative wins on realistic city-scale simulation.

17. QIRA — Quantum-Inspired Tensor-Network Optimization for Wireless Resource Allocation

Problem Statement: Joint power/channel/user-scheduling allocation in dense wireless networks (5G/6G small cells) is a combinatorially hard NP-hard optimization problem; existing heuristics or deep-RL approaches don't scale well or lack optimality guarantees.

Why existing solutions fail: Classical convex-relaxation methods (e.g., WMMSE) are slow to converge and get stuck in local optima for large networks; pure deep-RL scheduling lacks interpretability/convergence guarantees and requires extensive retraining per network topology.

Research Gap: No published method applies quantum-inspired tensor-network optimization (e.g., DMRG-style / Matrix Product State optimization, which classically simulates quantum annealing-style search) specifically to the joint power-and-scheduling allocation problem with a formal approximation-ratio analysis.

Proposed Innovation: Tensor-Network Resource Allocation (TN-RA) — a new algorithm that encodes the discrete scheduling + continuous power-allocation problem as a Matrix Product State (MPS) optimization (quantum-inspired, run classically), using a DMRG-style local-optimization sweep to find near-optimal joint allocations faster than exhaustive/branch-and-bound search, with a proven approximation bound under bounded interference graphs.

Technical Architecture: Simulated dense small-cell network (Python/MATLAB); TN-RA solver as a standalone optimization module benchmarked against WMMSE and deep-RL scheduling baselines.

Algorithms: New MPS-based DMRG-style combinatorial optimization algorithm adapted for wireless resource allocation (this is the core original contribution — a genuinely new optimization algorithm, not merely applying an existing tensor-network library); approximation-ratio proof under interference-graph sparsity assumptions.

Technologies: Python (NumPy, `quimb`) or custom tensor-network code — used only for tensor primitives, with the allocation algorithm itself being new), MATLAB for WMMSE baseline reproduction, PyTorch for deep-RL baseline.

Dataset: Not applicable (simulation-based); network topologies generated per standard 3GPP dense-urban small-cell simulation parameters.

Hardware: No special hardware; moderate cloud compute for large-network simulation sweeps.

Expected measurable improvement: $\geq 90\%$ of optimal (branch-and-bound ground truth on small instances) solution quality at $\geq 10\times$ faster runtime vs. WMMSE, and better scalability (handles $2\text{--}3\times$ larger networks in fixed time budget) vs. deep-RL baselines that require per-topology retraining.

Evaluation Metrics: Sum-rate/throughput achieved vs. optimal, runtime scaling with network size, convergence stability, generalization across topologies without retraining.

Difficulty: 9/10 **Research Novelty:** 9/10 **Implementation Complexity:** High (requires tensor-network math background) **IEEE Publication Potential:** High (IEEE Transactions on Wireless Communications, ICC, GLOBECOM) **Patent Potential:** Medium-High **Industry Demand:** Medium-High (5G/6G network optimization vendors) **Estimated Timeline:** M1-3 tensor-network theory + small-scale prototype; M4-6 full TN-RA algorithm + baselines; M7 large-scale evaluation + writeup. **Why judges will find it impressive:** "Quantum-inspired" algorithmic novelty applied classically to a concrete, high-

value telecom problem, with a formal approximation guarantee, is a rare combination for a capstone.

18. AquaRoute — Adaptive Depth-and-Salinity-Aware Routing Protocol for Underwater Acoustic Sensor Networks

Problem Statement: Underwater acoustic sensor networks (ocean monitoring, offshore infrastructure) suffer extremely high, spatially variable latency and packet loss because acoustic propagation speed/attenuation depends on depth, salinity, and temperature gradients that terrestrial-derived routing protocols ignore.

Why existing solutions fail: Adapted terrestrial protocols (e.g., depth-based routing like DBR) use only depth as a routing metric and assume roughly uniform acoustic channel quality, ignoring realistic sound-speed profile variation that causes shadow zones and unpredictable link quality even between geometrically close nodes.

Research Gap: No routing protocol incorporates a real-time-estimated local sound-speed-profile (derived from cheap onboard sensors) into the routing-cost metric, despite this being a first-order determinant of actual link quality underwater.

Proposed Innovation: Sound-Speed-Profile-Aware Routing (SSPAR) — a new geographic/opportunistic routing protocol where each node estimates a local channel-quality metric from onboard depth/temperature sensors (proxy for sound-speed profile via a simplified Mackenzie equation) and uses this in a modified void-avoidance greedy-forwarding algorithm, with a new recovery strategy for acoustic shadow zones.

Technical Architecture: Simulated underwater network (since real hardware is costly) using an established underwater acoustic channel simulator, extended with the new routing protocol; optional small-scale pool/tank hardware demo with off-the-shelf acoustic modems if budget allows.

Algorithms: New sound-speed-aware cost metric for geographic routing; new shadow-zone recovery/detour algorithm (extension of void-handling in geographic routing, but driven by acoustic-propagation prediction rather than pure geometry).

Technologies: Aqua-Sim (ns-2/ns-3 based underwater network simulator), Python for sound-speed-profile modeling (Mackenzie/UNESCO equations), optional low-cost underwater acoustic modem kits for a tank-scale demo.

Dataset: Public oceanographic sound-speed-profile datasets (e.g., World Ocean Atlas) for realistic environmental modeling in simulation.

Hardware: Optional: 2–3 low-cost acoustic modems (e.g., research-grade like AquaComm-class or DIY piezo-based) and a large water tank for a proof-of-concept demo; otherwise simulation-only is acceptable given budget.

Expected measurable improvement: $\geq 25\%$ improvement in packet delivery ratio and $\geq 20\%$ reduction in average end-to-end delay vs. DBR and VBF (Vector-Based Forwarding) baselines under realistic stratified sound-speed profiles in simulation.

Evaluation Metrics: Packet delivery ratio, end-to-end delay, energy consumption per delivered packet, void/shadow-zone recovery success rate.

Difficulty: 7/10 **Research Novelty:** 8/10 **Implementation Complexity:** Medium-High (mostly simulation-based, reducing hardware risk) **IEEE Publication Potential:** High (IEEE Oceanic Engineering, WUWNet, IoTJ) **Patent Potential:** Medium **Industry Demand:** Medium (offshore energy, ocean monitoring, defense) **Estimated Timeline:** M1-2 simulator setup + baseline protocols (DBR/VBF); M3-5 SSPAR design + implementation; M6 simulation campaign (+ optional tank demo); M7 writeup. **Why judges will find it impressive:** A genuinely underexplored niche (underwater networking) with real oceanographic data grounding and a physically well-motivated new metric.

19. MicroShard — A Lightweight Sharded Consensus Mechanism for High-Throughput IoT Microtransactions

Problem Statement: IoT devices increasingly need to record high-frequency micropayments/microtransactions (machine-to-machine energy trading, metered access) but existing blockchain consensus mechanisms are far too slow/energy-heavy for constrained devices and high transaction rates.

Why existing solutions fail: PoW is energy-prohibitive for IoT; existing sharded blockchains (e.g., Ethereum 2.0-style sharding) are designed for general-purpose smart contracts and assume relatively powerful validator nodes, not resource-constrained IoT devices with intermittent connectivity.

Research Gap: No consensus mechanism is co-designed for IoT-specific transaction patterns (small-value, high-frequency, geographically clustered) with a sharding scheme that minimizes cross-shard communication using transaction-locality-aware dynamic shard assignment.

Proposed Innovation: Locality-Aware Dynamic Sharding Consensus (LADS-C) — a new consensus mechanism combining lightweight BFT (PBFT-variant, since IoT clusters are small, ~10-50 nodes per shard) within shards, with a novel dynamic shard-reassignment algorithm that clusters nodes by observed transaction locality (who transacts with whom) to minimize expensive cross-shard atomic commits.

Technical Architecture: Simulated network of IoT nodes (or real Raspberry Pi cluster) organized into dynamically-formed shards; each shard runs lightweight BFT; a coordination layer handles cross-shard transactions and periodic re-sharding.

Algorithms: New transaction-locality-based dynamic shard-formation algorithm (community-detection-inspired but designed for online/streaming reassignment with bounded churn cost); lightweight BFT consensus tuned for small shard sizes; cross-shard atomic commit protocol optimized to minimize 2-phase-commit overhead.

Technologies: Go or Rust for the consensus implementation, Raspberry Pi cluster (or cloud VMs) for a real distributed deployment, simulated transaction workloads modeled on M2M energy-trading datasets.

Dataset: Synthetic microtransaction workloads calibrated against real M2M energy-trading pilot studies (e.g., peer-to-peer solar energy trading data where available in literature).

Hardware: 10–20 Raspberry Pi / low-cost SBC nodes (or cloud VM equivalent), which is realistic within a moderate budget.

Expected measurable improvement: $\geq 5\times$ higher transaction throughput and $\geq 80\%$ lower energy consumption per transaction vs. a PoW baseline, and $\geq 30\%$ reduction in cross-shard transaction latency vs. a static-sharding baseline, under realistic locality-clustered workloads.

Evaluation Metrics: Transactions per second, energy per transaction, cross-shard commit latency, shard-reassignment churn overhead, fault tolerance under simulated node failures.

Difficulty: 8/10 **Research Novelty:** 7/10 **Implementation Complexity:** High (distributed systems + real cluster deployment) **IEEE Publication Potential:** High (IEEE ICBC, Blockchain conference, IoTJ) **Patent Potential:** Medium-High **Industry Demand:** Medium-High (energy trading, supply chain IoT, machine economy) **Estimated Timeline:** M1–2 baseline PBFT + static sharding; M3–5 LADS-C dynamic sharding algorithm + cross-shard protocol; M6 real cluster deployment + benchmarking; M7 writeup. **Why judges will find it impressive:** A working multi-node real cluster demo processing live microtransactions with measurable throughput/energy wins is tangible and industry-relevant.

20. HAVA-NAS — Hardware-Aware Neural Architecture Search with a Learned Latency-Energy Co-Predictor for Embedded Vision

Problem Statement: Deploying CV models on diverse embedded hardware (different MCUs/NPUs) requires architectures tuned per-device, but running NAS with real on-device measurement for every candidate is prohibitively slow, and existing latency predictors are trained per-hardware-target with expensive data collection.

Why existing solutions fail: Hardware-aware NAS methods (e.g., ProxylessNAS, MCUNet) train separate latency predictors per target device from scratch, requiring thousands of on-device measurements per new hardware target, making them impractical for the long tail of embedded devices.

Research Gap: No NAS latency/energy predictor generalizes *across* hardware targets from a small number of calibration measurements on a new device, using transferable hardware-characteristic embeddings.

Proposed Innovation: Cross-Hardware Transferable Predictor NAS (CHTP-NAS) — a new NAS search algorithm guided by a **hardware-embedding-conditioned latency/energy predictor**: a model trained across many (architecture, hardware-descriptor, measured latency/energy) tuples from several devices, that generalizes to a new unseen device with only ~ 20 – 50 calibration measurements (few-shot), integrated into an evolutionary NAS search loop.

Technical Architecture: NAS search controller (evolutionary algorithm) + the new cross-hardware predictor (a GNN over the architecture graph, conditioned on a learned hardware-descriptor embedding) guiding architecture selection without exhaustive on-device runs.

Algorithms: New cross-hardware latency/energy predictor architecture (GNN + hardware embedding, trained via meta-learning/few-shot adaptation — e.g., MAML-style fine-tuning

on new-device calibration points); evolutionary NAS search loop using the predictor as a surrogate fitness function.

Technologies: PyTorch, NAS-Bench-201 or a custom mobile-search-space, actual embedded targets (e.g., STM32 + Raspberry Pi + Jetson Nano, to get genuinely diverse hardware for cross-device generalization), TensorFlow Lite Micro/TVM for on-device deployment and ground-truth measurement.

Dataset: CIFAR-10/ImageNet-subset for accuracy evaluation of found architectures; self-collected (architecture, hardware, latency/energy) measurement dataset across 3+ real embedded devices for training/validating the cross-hardware predictor.

Hardware: 3+ diverse embedded boards (STM32, Raspberry Pi, Jetson Nano — moderate total cost, ~\$150–250), power measurement tools.

Expected measurable improvement: $\geq 3\times$ reduction in on-device calibration measurements needed to reach predictor accuracy comparable to per-device-trained predictors (Spearman correlation ≥ 0.9 with real measurements), and NAS-found architectures achieving $\geq 95\%$ of hand-tuned baseline accuracy at target latency budget.

Evaluation Metrics: Predictor rank-correlation with ground-truth latency/energy, few-shot calibration sample efficiency, final found-architecture accuracy-vs-latency Pareto front vs. MCUNet/ProxylessNAS baselines.

Difficulty: 9/10 **Research Novelty:** 8/10 **Implementation Complexity:** Very High (NAS + multi-hardware deployment pipeline) **IEEE Publication Potential:** High (MLSys, CVPR efficient-DL workshop, DAC) **Patent Potential:** Medium-High **Industry Demand:** High (every embedded-AI vendor needs fast hardware-aware model tuning) **Estimated Timeline:** M1–2 search space + baseline NAS reproduction on 1 device; M3–5 cross-hardware predictor + multi-device data collection; M6 full CHTP-NAS search + Pareto evaluation; M7 writeup. **Why judges will find it impressive:** Solves a documented, expensive industry bottleneck (per-device NAS cost) with a genuinely transferable ML technique, validated on real diverse hardware.

21. AllWeatherFusion — Adversarial-Weather-Robust Multi-Sensor Fusion Architecture for Autonomous Vehicle Perception

Problem Statement: Autonomous vehicle perception (camera + LiDAR + radar fusion) degrades severely in adverse weather (fog, heavy rain, snow) because most fusion architectures are trained predominantly on clear-weather data and fail to adaptively reweight sensor trust when one modality becomes unreliable.

Why existing solutions fail: Fixed early/late-fusion architectures apply static fusion weights learned mostly from clear-weather-dominated datasets (e.g., KITTI, nuScenes clear-weather subsets); when camera is fogged out or LiDAR returns are corrupted by rain-scatter, these architectures don't dynamically down-weight the degraded modality, causing detection failures.

Research Gap: No fusion architecture explicitly estimates per-modality, per-region reliability *online* (without needing explicit weather labels) and uses this to dynamically reweight cross-modal attention in the fusion backbone.

Proposed Innovation: Reliability-Gated Cross-Modal Attention (RGCA) — a new fusion attention mechanism where a lightweight per-modality reliability estimator (trained via a self-supervised consistency signal between modalities) produces per-region confidence maps that gate a cross-attention fusion module, allowing the network to automatically down-weight a degraded sensor stream without explicit weather-condition labels.

Technical Architecture: Camera + LiDAR (+ optionally radar) backbone feature extractors → RGCA fusion module → detection head (e.g., built on a modified BEVFusion-style architecture, with the *fusion attention mechanism itself* being the new contribution).

Algorithms: New self-supervised cross-modal reliability estimation (consistency-loss-trained); new reliability-gated cross-attention fusion mechanism (core novel contribution — a new attention mechanism, not merely reusing existing fusion transformer blocks).

Technologies: PyTorch, nuScenes/Waymo Open Dataset (which include some adverse-weather samples) plus the CADC or DENSE (Seeing Through Fog) adverse-weather-specific datasets for stress-testing, mmdetection3d as an engineering base for the detection pipeline scaffolding (with the fusion module custom-built).

Dataset: nuScenes, Waymo Open Dataset, DENSE/"Seeing Through Fog" dataset (real fog-chamber-collected multi-sensor data — ideal for this project), CADC (Canadian Adverse Driving Conditions dataset, snow).

Hardware: Cloud GPU (2-4× A100/L4 equivalent) for 3D detection training, which is compute-intensive; this is the most GPU-hungry idea in the list and should be scoped/budgeted accordingly.

Expected measurable improvement: ≥15-20% mAP improvement on adverse-weather subsets (fog/rain/snow) vs. static-fusion baselines (e.g., standard BEVFusion), while maintaining comparable performance (≤2% drop) on clear-weather data.

Evaluation Metrics: 3D detection mAP (clear vs. adverse-weather splits), per-modality reliability-estimate correlation with ground-truth weather severity, inference latency.

Difficulty: 9/10 **Research Novelty:** 8/10 **Implementation Complexity:** Very High (3D perception + heavy compute) **IEEE Publication Potential:** High (CVPR/ICCV workshop, IROS, IEEE T-IV) **Patent Potential:** High (directly relevant to AV industry IP) **Industry Demand:** Very High (every AV company struggles with adverse-weather robustness) **Estimated Timeline:** M1-3 baseline fusion reproduction (heavy engineering lift); M4-6 RGCA design + training; M7 adverse-weather evaluation + writeup. (*Note: budget extra time/compute buffer — this is the most resource-intensive idea.*) **Why judges will find it impressive:** Tackles one of the most publicly recognized unsolved AV problems with a novel, interpretable (reliability-map-producing) mechanism and strong benchmark grounding.

22. GenoStream — A Reference-Free Streaming Compression Algorithm for Real-Time Nanopore Genomic Sequencing Data

Problem Statement: Real-time nanopore sequencing produces raw signal and basecalled read data at high rates in field/point-of-care settings (e.g., outbreak surveillance) with limited storage/bandwidth for transmission to analysis servers, but existing genomic

compressors are either reference-based (unavailable for novel pathogens) or too slow for real-time streaming.

Why existing solutions fail: Reference-based compressors (e.g., CRAM) need a known reference genome, unavailable for novel/emerging pathogens; general-purpose reference-free compressors (e.g., standard FASTQ gzip/zstd) don't exploit genomic-sequence-specific statistical structure and aren't designed for single-pass streaming with bounded memory.

Research Gap: No reference-free compression algorithm is designed specifically for streaming (single-pass, bounded-memory) operation on nanopore long-read data, exploiting long-read-specific repeat/error statistics (nanopore has distinctive error profiles vs. Illumina short reads).

Proposed Innovation: Streaming Context-Mixing Genomic Compressor (SCM-GC) — a new compression algorithm combining an online-updated, bounded-memory k-mer context model (exploiting long-read local repeat structure) with an arithmetic coder, plus a novel error-profile-aware model component that separately models nanopore's characteristic indel-heavy error pattern to improve compression of the quality-score stream (often the largest component of FASTQ files).

Technical Architecture: Streaming compressor library (C++) processing basecalled FASTQ (or raw signal, as a stretch goal) in a single pass with a fixed memory ceiling suitable for field laptops; benchmarked in a simulated real-time sequencing pipeline.

Algorithms: New bounded-memory online context-mixing model for nucleotide sequence compression (adaptive order-k Markov model with memory-bounded context table via hashing/eviction — a new data structure contribution); new nanopore-error-profile-aware quality-score compression model (separate context model for indel-heavy error patterns).

Technologies: C++ for the core compressor (performance-critical), Python bindings for pipeline integration, ONT (Oxford Nanopore) public FASTQ datasets for evaluation.

Dataset: Public nanopore sequencing datasets (e.g., ONT open datasets, NCBI SRA nanopore runs) covering bacterial/viral genomes relevant to outbreak-surveillance use cases.

Hardware: No special hardware; standard workstation sufficient; optional real MinION device if accessible (nice-to-have, not required since public datasets suffice).

Expected measurable improvement: $\geq 15\text{--}25\%$ better compression ratio vs. gzip/zstd baselines on both sequence and quality-score streams, at streaming throughput sufficient to keep pace with real-time MinION output rates ($\geq$ a few MB/s), with bounded memory ($< 1\text{GB}$).

Evaluation Metrics: Compression ratio, compression/decompression throughput (MB/s), peak memory usage, end-to-end streaming latency.

Difficulty: 7/10 **Research Novelty:** 7/10 **Implementation Complexity:** Medium-High (systems + information theory) **IEEE Publication Potential:** High (Bioinformatics journal, DCC — Data Compression Conference, IEEE/ACM TCBB) **Patent Potential:** Medium **Industry Demand:** Medium-High (field genomic surveillance, point-of-care diagnostics) **Estimated Timeline:** M1–2 baseline compressor benchmarking (gzip/zstd/existing FASTQ

compressors); M3-5 SCM-GC algorithm design + implementation; M6 large-scale dataset evaluation; M7 writeup. **Why judges will find it impressive:** Combines information theory, systems engineering, and genomics for a socially valuable (outbreak surveillance) application with clean, measurable compression-ratio results.

23. FedSync — A Bandwidth-Adaptive Synchronization Protocol for Staggered Distributed Edge Model Updates

Problem Statement: Fleets of edge devices (e.g., smart cameras, robots) running continually-updated ML models need to receive model updates efficiently, but heterogeneous, fluctuating network conditions across devices cause either excessive synchronization delay (waiting for slow devices) or wasted bandwidth (redundant full-model pushes).

Why existing solutions fail: Standard model-update distribution (full-model push, or naive delta-compression) doesn't adapt update granularity/scheduling per-device based on real-time bandwidth *and* model-staleness tolerance, and existing gossip-based model-sync protocols aren't designed for asymmetric edge/cloud topologies with heterogeneous device capability.

Research Gap: No synchronization protocol jointly optimizes (a) which model layers/deltas to send to which devices, (b) scheduling order, and (c) compression level, as a function of per-device bandwidth estimate and staleness tolerance, with a formal bound on worst-case fleet-wide staleness.

Proposed Innovation: Staleness-Bounded Adaptive Sync (SBAS) — a new synchronization protocol/algorithm that models the update-distribution problem as a bandwidth-constrained scheduling problem, solved with a new greedy-with-lookahead scheduling algorithm that selects per-device delta granularity (layer-wise vs. full) and compression level to minimize max fleet-wide staleness under a global bandwidth budget, with a provable staleness bound.

Technical Architecture: Central model-update server + fleet of simulated (or real, e.g., Raspberry Pi) edge clients; SBAS scheduler sits in the update-distribution path; supports layer-wise delta encoding.

Algorithms: New bandwidth-constrained scheduling algorithm (greedy-with-lookahead, with an approximation-ratio analysis relative to the optimal offline schedule); layer-wise delta significance scoring (which layers changed most, to prioritize transmission) — a new lightweight delta-prioritization metric.

Technologies: Python/Go for the sync server and scheduler, PyTorch for model delta computation, real Raspberry Pi fleet (10–15 nodes) or network-emulated cloud VMs with $\boxed{tc}$ -shaped heterogeneous bandwidth.

Dataset: Not applicable directly; uses standard vision model checkpoints (e.g., successive fine-tuning checkpoints of a MobileNet/YOLO model) as the "updates" being distributed, with realistic edge-fleet bandwidth traces (e.g., cellular/Wi-Fi throughput traces from public datasets).

Hardware: 10–15 Raspberry Pi nodes (or cloud VM equivalents), network emulation tools.

Expected measurable improvement: $\geq 30\%$ reduction in fleet-wide max staleness (time for all devices to reach a fresh model) at equal total bandwidth budget vs. round-robin full-model-push and naive delta-broadcast baselines.

Evaluation Metrics: Max/average fleet staleness, total bandwidth consumed, per-device model accuracy over time (as a function of staleness), scheduling algorithm runtime overhead.

Difficulty: 7/10 **Research Novelty:** 7/10 **Implementation Complexity:** Medium-High **IEEE Publication Potential:** Medium-High (IEEE Edge computing venues, INFOCOM workshop, IoTJ) **Patent Potential:** Medium **Industry Demand:** High (fleet ML-model management is a real operational challenge for edge-AI companies) **Estimated Timeline:** M1-2 baseline sync methods + fleet testbed; M3-5 SBAS scheduling algorithm + delta-prioritization; M6 real-fleet evaluation; M7 writeup. **Why judges will find it impressive:** A concrete, provably-bounded scheduling algorithm applied to a real fleet demo (visible dashboard of staleness dropping) is a clean, convincing systems result.

24. SpecEdge — Hardware-Co-Designed Speculative Decoding for LLM Serving on Memory-Constrained Edge GPUs

Problem Statement: Speculative decoding (using a small draft model to propose tokens verified by a larger model) speeds up LLM inference on datacenter GPUs, but on memory-constrained edge GPUs (e.g., Jetson-class, consumer laptops), keeping both draft and target models resident, plus KV caches for both, often exceeds available memory, negating the benefit.

Why existing solutions fail: Standard speculative decoding implementations assume ample GPU memory to hold both models fully; naive quantization of the draft model to save memory increases draft-rejection rate, eroding speedup; no existing method jointly optimizes draft-model quantization level and speculation length as a function of real-time memory pressure and acceptance-rate feedback.

Research Gap: No adaptive speculative decoding scheme dynamically tunes both the draft model's quantization/precision *and* the speculation-window length online, based on a live estimate of the accuracy/memory/speed tradeoff, specifically for memory-constrained single-GPU edge deployment.

Proposed Innovation: Memory-Adaptive Speculative Decoding (MASD) — a new decoding algorithm that maintains a small controller estimating current token-acceptance-rate and available memory headroom, and dynamically adjusts (a) draft-model activation precision (via a novel mixed-precision draft-execution scheme that swaps precision per-layer without model reload) and (b) speculation length, to maximize expected tokens/sec under a strict memory ceiling — this joint dynamic control policy is the new contribution (not merely applying existing speculative decoding).

Technical Architecture: Runs atop a HuggingFace/vLLM-style serving stack on a single edge GPU (e.g., Jetson Orin or a laptop RTX GPU); custom CUDA/Triton kernels for fast per-layer precision switching without full model reload.

Algorithms: New online controller (lightweight bandit/control-loop) jointly tuning quantization precision + speculation length; new low-overhead per-layer precision-

switching kernel design (memory-management contribution).

Technologies: PyTorch, HuggingFace Transformers, custom Triton/CUDA kernels, small LLaMA-family models (e.g., 1B draft / 7-8B target) for realistic experimentation within GPU budget.

Dataset: Standard LLM evaluation sets (e.g., MT-Bench, AlpacaEval subset) for measuring both speed and output-quality preservation.

Hardware: 1 edge-class GPU (e.g., Jetson Orin 32/64GB or a single consumer RTX GPU with limited VRAM, e.g., 8-12GB, to genuinely stress memory constraints).

Expected measurable improvement: $\geq 25\text{--}35\%$ higher tokens/sec vs. static (fixed-precision, fixed-speculation-length) speculative decoding baselines under the same strict memory budget, with negligible ($\leq 1\%$) output-quality degradation vs. full-precision baseline.

Evaluation Metrics: Tokens/sec throughput, peak GPU memory usage, token-acceptance rate, output quality (perplexity/win-rate on eval benchmarks).

Difficulty: 8/10 **Research Novelty:** 8/10 **Implementation Complexity:** High (CUDA/kernel engineering) **IEEE Publication Potential:** High (MLSys, EMNLP systems track, ACL industry track) **Patent Potential:** Medium-High **Industry Demand:** Very High (on-device/edge LLM serving is a hot, resource-constrained problem) **Estimated Timeline:** M1-2 baseline speculative decoding reproduction; M3-5 MASD controller + precision-switching kernels; M6 edge-GPU benchmarking; M7 writeup. **Why judges will find it impressive:** Directly tackles the practical, widely-felt pain of running capable LLMs on constrained hardware with real kernel-level engineering and measurable throughput gains.

25. GEMFusion — Multimodal Gaze-and-EMG Fusion Architecture for Low-Latency Assistive Device Control

Problem Statement: Assistive control interfaces for users with severe motor impairments (e.g., ALS, high spinal cord injury) using single-modality control (eye-tracking alone or EMG alone) suffer from either fatigue (prolonged gaze-based dwell selection) or unreliability (noisy residual EMG signals), limiting practical usability for fine-grained device control (e.g., wheelchair/robotic arm).

Why existing solutions fail: Gaze-only interfaces require dwell-time selection causing "Midas touch" errors and eye fatigue; EMG-only interfaces suffer signal drift and cross-talk from residual muscle noise in severely impaired users; naive late-fusion (simple voting) of the two modalities doesn't exploit their complementary temporal characteristics (gaze indicates *intent target*, EMG indicates *confirmed action timing*).

Research Gap: No published fusion architecture explicitly models the complementary temporal roles of gaze (spatial intent) and EMG (temporal confirmation) via a structured cross-modal representation, rather than naive concatenation or majority voting.

Proposed Innovation: Intent-Confirmation Cross-Modal Fusion (ICCMF) — a new architecture with two specialized sub-networks (a gaze-based spatial-intent predictor and an EMG-based temporal-confirmation detector) combined via a novel **gated cross-modal fusion layer** that uses gaze-predicted intent to contextually gate the EMG confirmation

threshold (rather than treating both streams symmetrically), reducing false positives from EMG noise while cutting gaze dwell-time requirements.

Technical Architecture: Eye-tracker (e.g., Tobii or webcam-based gaze estimation) + surface EMG electrodes (e.g., on a residual-functional muscle group) feeding a real-time fusion pipeline controlling a simulated (or physical, if available) robotic arm/wheelchair interface.

Algorithms: New gated cross-modal fusion layer (the core novel architecture contribution — a new attention/gating mechanism specific to asymmetric-role multimodal fusion); new feature-extraction pipeline for low-latency gaze-intent classification and EMG-onset detection.

Technologies: PyTorch, OpenCV/MediaPipe or Tobii SDK for gaze tracking, low-cost EMG sensor (e.g., Myoware), Robot Operating System (ROS) for controlling a demo robotic arm.

Dataset: Self-collected gaze+EMG session data from able-bodied volunteers simulating impaired-control scenarios (IRB-lightweight, standard for HCI capstones) for training/validation; public gaze-estimation datasets (e.g., GazeCapture) for pretraining the gaze sub-network.

Hardware: Webcam or Tobii eye-tracker (~\$150-1000 depending on budget tier), Myoware EMG sensor kit (~\$40), a small robotic arm kit (~\$150-300) for the demo.

Expected measurable improvement: $\geq 30\%$ reduction in average selection time (vs. gaze-only dwell-based baseline) and $\geq 40\%$ reduction in false-positive action triggers (vs. EMG-only baseline), measured in a controlled target-selection task with volunteer participants.

Evaluation Metrics: Selection time, false-positive/false-negative action rate, information transfer rate (bits/min, standard HCI/BCI metric), user-reported fatigue (subjective survey).

Difficulty: 7/10 **Research Novelty:** 7/10 **Implementation Complexity:** Medium-High **IEEE Publication Potential:** High (IEEE TNSRE, CHI, ASSETS — ACM accessibility conference) **Patent Potential:** Medium **Industry Demand:** Medium-High (assistive tech is a growing, socially impactful market) **Estimated Timeline:** M1-2 sensor pipelines + baseline single-modality systems; M3-5 ICCMF fusion architecture + training; M6 user study (target-selection task, N $\approx$ 10-15 volunteers); M7 writeup. **Why judges will find it impressive:** A live, empathetic assistive-tech demo controlling a physical robotic arm via gaze+muscle signals, backed by a genuinely new fusion mechanism and a small human-subject user study, is memorable and impactful.

RANKING

Scored 1-10 (except Feasibility, which is qualitative) across: **Novelty, Research Value, Industry Impact, Patentability, Difficulty (lower engineering-risk = higher score), Feasibility in 7 months, IEEE Publication Probability, Future Startup Potential.**

Rank	Idea	Novelty	Research Value	Industry Impact	Patentability	Difficulty (risk-adj.)	7-Month Feasibility	IEEE Pub. Probability	Stability Potential
1	#4 EvictAttn (LLM KV-cache eviction)	8	9	10	8	7	High	9	9
2	#24 SpecEdge (edge speculative decoding)	8	9	10	7	7	High	8	9
3	#3 FedCluster-NC (federated clustering)	8	8	9	6	7	High	9	7
4	#20 HAVA-NAS (cross-hardware NAS)	8	8	9	7	6	Medium-High	8	8
5	#21 AllWeatherFusion (AV adverse-weather)	8	9	10	8	5	Medium (compute-heavy)	9	8
6	#7 SwarmLoc (GPS-denied drone swarm)	8	8	8	7	5	Medium	8	8
7	#6 LightPQ (lightweight PQC for IoT)	8	8	8	8	5	Medium	8	6
8	#17 QIRA (quantum-inspired wireless opt.)	9	8	7	7	5	Medium	8	6
9	#10 HAOS-Sched (heterogeneous OS scheduler)	8	8	8	7	4	Medium (kernel risk)	8	6
10	#14 DP-FedMed (adaptive DP budget)	8	8	8	6	6	High	8	6
11	#9 SparseRV (RISC-V sparse compiler)	8	8	7	6	4	Medium (compiler risk)	8	6
12	#13 NeuroDenoise (EEG BCI)	8	7	6	6	6	High	7	6

Rank	Idea	Novelty	Research Value	Industry Impact	Patentability	Difficulty (risk-adj.)	7-Month Feasibility	IEEE Pub. Probability	Stability Potential
	denoising)								
13	#5 LEO-CC (satellite congestion control)	7	8	9	7	5	Medium (kernel risk)	8	7
14	#25 GEMFusion (gaze+EMG assistive)	7	7	7	6	7	High	7	7
15	#2 FlickerFree-LiFi (adaptive VLC)	8	8	6	8	5	Medium (analog risk)	8	5
16	#1 FloodMesh (disaster mesh comms)	7	7	8	5	7	High	7	6
17	#19 MicroShard (sharded IoT consensus)	7	7	6	7	5	Medium	7	6
18	#8 EdgeSched (deadline-aware CV offload)	7	7	8	5	7	High	7	6
19	#12 IntermitConsensus (intermittent-link DB)	7	7	6	5	6	Medium-High	7	5
20	#11 PerceptABR (perceptual ABR streaming)	7	7	8	5	7	High	7	6
21	#16 EvacSwarm (evacuation traffic MARL)	7	7	6	3	5	Medium	7	3
22	#23 FedSync (fleet model sync)	7	6	7	5	7	High	6	6
23	#18 AquaRoute (underwater routing)	8	7	5	5	7	High (sim-only OK)	7	4

Rank	Idea	Novelty	Research Value	Industry Impact	Patentability	Difficulty (risk-adj.)	7-Month Feasibility	IEEE Pub. Probability	Sta Pot
24	#22 GenoStream (nanopore compression)	7	7	6	5	7	High	7	4
25	#15 TinyStruct (SHM compressive sensing)	7	6	6	5	6	Medium-High	6	4

Top 10 Highlighted

1. **#4 EvictAttn** — highest combined feasibility, novelty, and industry relevance; the single strongest choice if the team wants maximum publication + startup upside for moderate hardware needs.
2. **#24 SpecEdge** — same "hot" LLM-serving space as #4, slightly more kernel-engineering risk but very strong industry pull.
3. **#3 FedCluster-NC** — deep ML-research contribution, fully simulation-based (low hardware risk), strong healthcare-FL story.
4. **#20 HAVA-NAS** — ambitious but very "MLSys-paper-shaped"; needs disciplined scoping of the multi-device data collection phase.
5. **#21 AllWeatherFusion** — highest industry impact ceiling (autonomous vehicles) but the most compute-hungry; only pick if the team can secure serious GPU budget/time.
6. **#7 SwarmLoc** — best "wow factor" live demo (multi-drone swarm); moderate hardware risk but very rewarding if it works.
7. **#6 LightPQ** — rigorous applied-crypto project with a clean, defensible original contribution and strong patent angle.
8. **#17 QIRA** — the most mathematically novel entry; ideal for a team with strong theory/math backgrounds, lower on immediate startup potential.
9. **#10 HAOS-Sched** — elite systems credibility (Linux kernel scheduling) but carries real kernel-development risk; scope the shim carefully.
10. **#14 DP-FedMed** — timely (healthcare AI + privacy regulation), formally rigorous, fully simulation-based so low hardware risk.

General advice for choosing: if the team is strongest in **ML systems / infra**, pick from {#4, #24, #20, #3}. If strongest in **embedded/hardware + robotics**, pick from {#7, #6, #21}. If strongest in **theory/algorithms**, pick {#17, #9}. If strongest in **networking/distributed systems**, pick {#5, #10, #12}. Matching the team's actual strengths to the idea is the single biggest determinant of finishing a *working, measurable, publishable* result in 7 months.